Executive Summary

Echoes of Marsili is an interactive narrative driven museum project that brings the life and discoveries of Luigi Ferdinando Marsili (1658 to 1730) into life. Hosted in Bologna's Palazzo Poggi Museum - Marsili Collection, the experience transforms static exhibits into a guided story: as visitors can interact with key artifacts, they go through the successive chapters of a Twine-based tale narrated by Marsili himself. This hybrid onsite experience blends history and science with gamified exploration, aiming to engage modern audiences in heritage learning. Key outcomes include a structured user journey through the museum, accessible design (audio, subtitles, multi-language), and a creative linkage between Marsili's archives and an interactive app. The following sections outline the museum context, narrative experience, technical design, and project details (including personas, tools, and implementation plans).

HOME

Echoes of Marsili

An Interactive Hybrid Narrative Experience for Heritage, Science, and Discovery

Meta Description: Echoes of Marsili is an interactive museum experience at Bologna's Palazzo Poggi - Marsili Collection. Visitors explore Luigi Ferdinando Marsili's scientific legacy through narrative storytelling triggered by digital multimedia.

The reflection of Marsili project reimagines a museum visit as a journey through history. By combining physical exhibits with a smartphone based story app, it turns Luigi Ferdinando Marsili's own collection into a living narrative. Visitors begin in the halls of the Palazzo Poggi Museum (Marsili Collection) and, Twine-based interactive narrative, unlock chapters of Marsili's life from his childhood to his pioneering oceanographic research. Each chapter unfolds as an immersive Twine story - complete with images, audio, and interactive questions that creates a personalised exploration of science and heritage. Result is a seamless blend of education and entertainment: users learn through Marsili's voice, making sense of complex topics (wars, currents, instruments) in an engaging way.

THE MUSEUM

Marsili Context:

Luigi Ferdinando Marsili (1658–1730) was a Bolognese nobleman, soldier and scientist who founded the city's Institute of Sciences (Istituto delle Scienze) in 1711. His passion for natural history, physics and exploration shaped the early collections of what became the Palazzo Poggi Museum. Under Marsili's vision, the palace at Via Zamboni became a hub for experimentation and learning. In fact, historical records note that Marsili convinced Bologna's leaders to transform Palazzo Poggi into a science academy in the early 18th century. Today, the Marsili Collection occupies refurbished rooms of the Palazzo Poggi Museum, permanently opened to the public in 2026.

Historical Background

This museum assembled one of Europe's richest scientific libraries and object collections. Marsili served in armies, travelled from Venice to Constantinople, and studied oceans, rivers and artillery. His legacy includes early oceanographic observations (notably of the Bosporus currents) and *Danubius Pannonico-Mysicus*, a detailed six-volume study of the Danube River. The Museum's Marsili wing traces this legacy: from naval models and maps he used, to laboratory instruments he invented (like a glass ampoule to measure water salinity). Marsili's story also involves dramatic episodes (battle, captivity, introducing coffee to Europe), many of which are echoed in the artifacts and exhibits on display. By connecting Marsili's biography to these objects, the museum underscores its founding mission: marrying war, science, and curiosity in one educational narrative.

Star Assets

Asset Name	Room	Significance
Historic Ship Model	Geography & Nautics	A 17th-18th century warship model (from Marsili's institute) that illustrates the naval technology behind his Bosporus experiment.
Fortress & Artillery	Military Architecture	Scale models of forts and cannons, reflecting Marsili's military career. He studied and sketched Ottoman artillery, which is represented here.
Ampoule Instrument	Physics Lab (Science Room)	A slender glass hydrometer, similar to one Marsili used to measure seawater density. This early oceanographic tool embodies his scientific method.
Danube River Map	Cartography/Map Room	A large engraved map from *Danubius Pannonico-Mysicus* (1726). It shows the Danube's course and marks Marsili's extensive river surveys.
Coral Specimen	Natural History Room	A fossilized coral (once deemed a plant). Marsili's studies helped prove coral are animals, and this specimen symbolizes his oceanographic discoveries.
Coffee Brewing Set	Cultural Collection (or Café)**	A recreated 17th-18th century coffee set. Marsili learned coffee preparation during his Ottoman captivity and later introduced it to Europe, making this a cultural touchstone of his story.
Marsili's Bust	Military Architecture	The marble bust of Luigi Marsili, installed in the museum. It marks the entrance to his world, literally putting Marsili's face in the narrative (source: museum photography).

Table: Key artifacts in the Marsili Collection (room and importance).

Institutional Goals

The Marsili Museum pursues its long-term goals through preservation, research and engagement. Core aims include:

Preservation & Growth: One of the primary goal is expand and care for scientific collections (instruments, maps, manuscripts) via acquisitions and donations.
Education & Outreach: Next is, making the collection accessible for public and students through interactive learning.
Research & Collaboration: Partnering with universities and laboratories for scientific studies
Public Engagement: Increase visitor numbers and participation, particularly by revitalising displays with modern interpretation (as we do with *Echoes of Marsili*).
International Profile: Globalise the collection or use the Marsili Collection to highlight Bologna's role in early modern science, attracting global scholars and tourists.

Weak Points

Currently, several opportunities for improvement stand out:

Limited Digital Interaction: The existing exhibits rely mostly on text panels. There are few multimedia or interactive elements to engage today's tech-savvy visitors.
Accessibility Gaps: Some galleries have few accommodations for impairments (e.g. limited audio guides or sign language). Physical layout and text size can be challenging for visitors with mobility or vision issues.
Static Presentation: Many displays are static; younger audiences or families may lose interest without activity.
Under-Communicated Context: Visitors without previous knowledge of 18th-century science may miss connections between artifacts and Marsili's story.
Fragmented Narrative: Story of Marsili journey spans many fields (military, geography, natural history) which can feel disjointed. So, we must weave these into a cohesive story, which is *Echoes of Marsili*'s goal.

THE EXPERIENCE

Main Concept

Echoes of Marsili is designed as a **concept-driven interactive narrative**. Core concept of Marsili museum based on exploration and curiosity, rather than passive panel visitors "converse" with Marsili through active participation: this project's Twine storyline uses Marsili's first-person voice to make history personal. We juxtapose authentic detail with mystery (for example, a coffee aroma hinting at the next story beat) to engage emotions and intellect. The goal is to evoke **wonder** at scientific discovery: whether it's uncovering hidden currents or debunking myths of sea monsters, each chapter sparks the desire to know more. In practice, this means blending multimedia (audio clips of Marsili reading, archival images, animations) with game-like tasks. This game encourages visitor to participate rather than just read, aligning with marsili moto "Learn through doing".

User Journey

Onsite Flow: Museum journey begin with the entrance of the Marsii collection inside Palazzo Poggi Museum, where the atmosphere immediately introduces them to Marsili's world through subtle ambient sound, historical visuals, and the feeling of entering an eighteenth-century study. Experience start with the short introductory prologue where Marsili himself welcome the visitor. From here, visitor then move through different thematic spaces connected to key moments of Marsili's life including ,travels, scientific discoveries, captivity and oceanographic research. As they go through the museum, each location unlocks a new chapter of the interactive Twine story on their device, letting the real environment and digital story unfold together. This flow is made to feel simple, immersive, and smooth, encouraging visitors to join in and explore the museum instead of just observing it passively.

The flow is roughly: **Entrance Hall → Marsili's Study (Chapter I) → Geography & Nautics (II) → Military Architecture (III–VIII) → Cartography/Map Room (IX–X) → Science Lab and Natural History (XI–XIII) → Exit Hall (XIV).** The sequence is linear (each chapter unlocks the next), ensuring a clear progression.

```
mermaid
```

graph LR
A[Entrance Hall: Story Prologue] --> B[Marsili Study (QR scan) → Chapter I: "Boy in Bologna"]
B --> C[Geography & Nautics Room → Chapter II: "Two Currents"]
C --> D[Military Architecture → Chapters III–VIII (The Rába, Captivity, Coffee, Cannons)]
D --> E[Cartography/Maps Room → Chapters IX–X (Return & Danube)]
E --> F[Physics & Natural History → Chapters XI–XIII (Oceanography)]
F --> G[Exit Hall → Chapter XIV: "The Gift" (Conclusion)] Diagram: A high-level user flow through museum areas and corresponding story chapters.

Narrative Exploration

The Twine script is divided into 14 chapters, each tied to an exhibit:

- **Chapter I – The Boy in Bologna:** Triggered at the Museum's entrance/study area, this covers Marsili's early life and rebellion against academic dogma.
- **Chapter II – Two Currents:** Activated in the Geography and Nautics room, it dramatizes Marsili's Bosporus experiment, linking to naval models and sea maps.
- **Chapters III–V – The Rába & Captivity:** In the Military Architecture room (with cannon models and battle schematics), these chapters recount Marsili's wartime experience, his near-death at the Rába River, and subsequent capture.
- **Chapters VI–VII – The Brothers & A Small Cup:** As visitors examine the Ottoman-era coffee brewing set, Marsili describes being sold to two Bosnian brothers and learning to make coffee, leading to the birth of his first book on coffee.
- **Chapter VIII – The Dangerous Pages:** Also in Military Architecture, triggered by views of artillery drawings, Marsili reveals his secret notes on Ottoman cannon technology written during captivity.

- **Chapters IX–X – The Long Way Home & The River, Drawn:** In the Cartography/Map Room, visitors trigger the story of Marsili's return to war and his survey of the Danube, directly relating to the river maps on display.
- **Chapters XI–XIII – The Ampoule, The Wet World & No Monsters:** In the Science and Natural History rooms, guests learn about Marsili's ocean measurements (using the ampoule instrument) and his discovery that coral reefs are animals, not monsters, tying into specimens exhibited.
- **Chapter XIV – The Gift: Back** In the final hall, Marsili explains that after his disgrace he donated all his collections to found the Istituto delle Scienze (the Museum's ancestor), asking visitors to "open the cases" and continue his mission.

Each chapter is a short conversation with Marsili and often ends with a fun interactive activity such as quiz and activity. Solving these reinforces the content and encourages exploration.

Hybrid Interaction

The experience is combination of physical museum with digital storytelling in a engaging and easy to use way. Each visitors go through the Palazzo Poggi Museum and the Marsili collection, where they interact with different marsii echo's like manuscripts, maps, scientific instruments, and historical objects that connect directly to different moments in Marsili's story. Visitors can open the Twine-based experience on their phones or tablets without downloading anything or needing technical skills. Story is develops through conversations and historical images, ambient sounds and small interactive activities that make visitor to make participate rather of reading information. The interface was designed and prototyped in "Figma" with a focus of clear navigation and accessibility, so even first-time users can easily explore the experience while staying connected to the museum and its heritage content.

Accessibility

Echoes of Marsili is designed to be accessible for different visitors. All narration includes subtitles, and audio can be adjusted or turned off. An Italian/English toggle helps both local and international visitors follow the story. The Twine experience also supports screen readers, while spoken dialogue and important sounds are provided in text form for hearing-impaired visitors. Some exhibits include tactile replicas, allowing visitors to explore objects through touch and making the experience more inclusive and engaging.

Learning Goals

This design is concept-oriented with a focus on *curiosity*, *authenticity*, and *active learning*. Key learning outcomes include:

- **Extending Knowledge:** Each Visitors will better understand Marsili's contributions to science and engineering (such as oceanography and artillery), seeing how data and experiment shaped his insights.
- **Stimulating Curiosity:** By exploring surprising ideas and completing small challenges, visitors are encouraged to think critically and ask questions, much like Marsili did in his own research.
- **Promoting Understanding through Experience:** Instead of only reading static information panels visitors explore historical moments like battle's, laboratory discovery.
- **Fostering Cultural Empathy:** Through the good story writhing and twine, (e.g. Marsili's treatment by Ottoman hosts) user experienced history beyond their own prespective.
- **Encouraging Self-Reflection:** Marsili often addresses the player directly, prompting reflection (for instance, on leadership or gender bias, as Caterina's brother had dismissed her). This personal connection reinforces emotional engagement.

Together, these goals align with Marsili's legacy: encouraging discovery through first-hand investigation, and appreciating science as part of human experience.

THE PROJECT

Personas

Two representative user personas guided our design:

- **Alessio (24, Student)** : An "Interactive Narrative Explorer." A humanities student from Bologna who loves history and interactive media. Alessio enjoys museum experiences that are *participatory* rather than passive. He gets bored by heavy text panels but is fascinated by storytelling and puzzles. Alessio's goals are to explore Marsili's world through the story, engage with multimedia quizzes, and see science as an adventure. He dislikes static exhibits and appreciates clear, dynamic content (e.g. audio, images).
- **Sara (24, Accessibility Advocate)** : An "Inclusive Experience Explorer." A visitor with visual impairment, Sara often visits museums but finds exhibits too visually oriented. She relies on audio, large-print text, and tactile models. Sara's goals are to navigate the experience independently, listen to Marsili's narration, and participate fully in quizzes. She values clear audio descriptions, subtitles, and straightforward navigation. Sara's frustrations include small text, poor layout, and any digital interface without screen-reader support.

These personas ensure the experience is inclusive and engaging: Alessio motivates features like branching narrative and quizzes, while Sara drives our accessibility solutions (audio narration, captions).

Twine Experience

The narrative was authored in **Twine** (v2) using its **SugarCube** format. We structured each chapter as a Twine passage, linking passages to simulate the user's choices. To connect museum exhibits, we created **unique passage IDs**: each QR code points to a Twine URL ending in a hash (e.g. `#TwoCurrents`) that opens directly to the corresponding passage. Twine's simple logic allowed us to implement conditional flows (e.g. recording which chapters are completed to show progress) and integrate JavaScript for minor interactive widgets (such as matching games or timers). All text, images and audio are packaged in the story HTML, which we host via **GitHub Pages**. This means the entire experience can run in a browser (online or cached offline).

Gameplay Structure

The experience is primarily narrative but uses light gamification. The Twine story is mostly linear (each chapter unlocks the next) with a few branching prompts (e.g. the choice between "Books vs Experience" shaping Marsili's response tone). To engage users, we embedded small tasks in some chapters:

- **Quizzes and Polls:** Simple questions (true/false or multiple choice) based on the story content. E.g., after "Two Currents" the app asks whether the Bosporus has opposing currents, or after "No Monsters" whether the deep sea has fish or serpent myths. Correct answers provide a pop-up "CORRECT!" message.
- **Puzzles:** Mini activities like dragging labels onto a map of the Danube, or identifying an instrument sound from a short audio clip (representing the Ampoule's beep).
- **Collectibles:** Each completed chapter earns the user a "digital stamp" or badge shown in the app. Collecting all 14 unlocks a final reward (see below).

The story choices are easy so as not to frustrate, and all quizzes are optional—the narrative continues even if the visitor skips a task. This ensures accessibility: anyone can proceed at their own pace. The overall structure is user-paced and exploratory, rather than competitive.

Prototype & Deliverables

Key artifacts and tools produced for the project include:

Artifact	Format	Tool(s)
Twine Interactive Story	Web App (HTML)	Twine
Website Prototype (Landing/Flow)	Responsive HTML/CSS/JS	Figma (design) + GitHub Pages (hosting)
UX/UI Mockups	Image/PDF	Figma
Persona & Research Docs	PDF	(Compiled from Figma/Word)
Gamification Flowcharts	Image	Figma + Mermaid
Final Report / Design Brief	Document (PDF)	(Various)

Table: Project deliverables, formats, and tools.

Interaction Design

We applied a human-centred design (PACT) framework to ensure the experience fits its context:

People: Our personas (Alessio, Sara) represent curious learners of varying abilities. The interface uses simple language and icons, and provides immediate feedback to keep both scholars and casual visitors engaged.
Activities: The main activity is "Quest-style exploration": scanning codes, reading/listening to Marsili, and answering questions. We designed each step to be short (1–2 minutes of content) so visitors can absorb information without fatigue. Optional game tasks are well-defined with minimal steps.
Context: This unfolds in a calm museum environment (air-conditioned, with ambient noise). Visitors may be alone or in groups. We assume typical museum hours, with moderate crowds. Since the museum may not offer robust mobile reception, the design allows offline operation (the Twine story is self-contained once loaded). Lighting in the historic rooms is low – so our app uses high-contrast visuals and adjustable brightness.
Technologies: The Twine story runs in any modern mobile browser (no special app needed). We made sure the experience works smoothly on both with Android and iOS. project is built using standard HTML and CSS and is hosted on GitHub Pages. Figma prototypes guided our UI layout, and we tested on multiple screen sizes.

Gamification Elements

Key gamification aspects include:

Emotional Engagement: Narrative (first-person voice of Marsili) creates personal connection between the visitor and the story. Dramatic elements like the smell of coffee hinting at a secret and battle chaos sounds in the headphone add sense of wonder and suspense.
Interactive Learning: Actively participation and small challenges turn passive reading into enjoyable, and exciting experience. By clicking, matching, or answering, users participate rather than memorize. This hands-on approach increases retention and satisfaction.
Progress Feedback: A progress meter or "stamp album" visible in the app shows how many chapters are complete, motivating visitors to continue.
Rewards: Completing all chapters yields a virtual reward (for example, a "Founding Member" digital certificate that can be emailed). On-site completers can show a staff member to receive a small physical memento (like a bookmark with Marsili's portrait).
Sensory Stimulation: Multi-sensory elements such as Marsili's audio narration, background music, and ambient sounds make each chapter memorable and immersive.
Challenge and Mastery: Quizzes and puzzles provide enough challenge to feel engaging. Because they are optional, users do not get stuck. Also, satisfaction of solving a puzzle or answering correctly reinforces learning and curiosity.

Constraints & Solutions

Key anticipated issues and our responses:

Connectivity: The museum wifi may not always be reliable. *Solution:* The Twine app is lightweight and preloads critical content. It can run offline once loaded.
Device Variability: Visitors may have old phones. *Solution:* We optimized the app for basic browsers and include large tap targets. We also provide a few museum tablets with the app preloaded.
Noise & Environment: Users might not hear audio clearly. *Solution:* Ensuring text transcript of all audio.
Accessibility: Different visitors may have different accessibility needs, including vision, hearing, or mobility support. *Solution:* we implemented subtitles, text resizing, alt-text for images, and a clearly structured UI. The museum also provide ramps and an alternative entrance for wheelchair users, while the app supports accessibility with large buttons and easy navigation for comfortable use.

Maintenance: Technology can sometimes fail. *Solution:* So, we chose stable and reliable tools such as Twine and a static website structure.
Engagement Risk: Few visitors might ignore the interactive parts of the experience. *Solution*: To encourage participation, staff members and signs at the entrance will draw attention to the QR codes and storyline. In Introduction scene, which includes sound and scent elements, is also designed to capture visitor's interest right from the beginning.

TEAM

Samuele – *Project Coordinator & Narrative Design.* Leads content research and authored the Marsili storyline.
Alpamy – *UX/UI & Visual Design.* Designed the app interface, pages and iconography in Figma; ensured accessible layouts.
Gaurav – *Twine Developer & Technical Integration.* Implemented the interactive story in Twine, set up QR linking, and managed GitHub hosting.
Mayank – *Interactive Prototyping & Gamification.* Developed the mini-quizzes and puzzles, and created prototype flows in Figma and Mermaid diagrams.

The team comprises Master's students in Cultural Heritage & Interactive Design at the University of Bologna (2025-2026).

Clarifying Questions: Should the narrative include any additional interactive elements (e.g. augmented reality overlays) beyond the current QR/Twine design? Are there specific visitor rewards or grading systems we should incorporate? Are further visual branding guidelines (colors, fonts) preferred for consistency with the Palazzo Poggi Museum identity?